Department of Computer Sciences

FACULTY OF COMPUTING AND ARTIFICIAL INTELLIGENCE

Programming of Fundamentals Project Lab Report

Topic: Gene Expression File Analyzer

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Introduction

This project report explains the design and implementation of a "**Gene Expression File Analyzer".** The main idea of the project is to create a simple console-based system that can store and manage information about gene expression for diseased patients and normal (healthy) samples. The program is written in C++ and focuses on basic concepts such as variables, functions, decision making, loops, and especially file handling, as required.

In real life, biologists measure how strongly a gene is expressed in different samples. High or low expression levels can give clues about diseases. In this project, that real concept is turned into a small software system. The system allows the user to **add records, view all records, search for a specific record, edit an existing record, and delete a record from the files.** All data is stored in simple text files so that it remains available even after closing the program.

Problem statement and objectives

The given assignment asked to implement an entity from a project list with at least five variables for each record. Five operations had to be supported: **Add, Edit, View, Search, and Delete.** Furthermore, all records needed to be stored in files instead of only in memory.

To meet these requirements, the chosen entity was "gene expression sample". Two types of samples are handled: **diseased patient samples and normal samples**. For each sample, the program stores the following five fields:

- Sample ID
- Person Name
- Age
- Gene Name
- Gene Expression Level

The main objectives of the project are:

- To practice using variables, input and output statements, and conditional statements in C++.
- To learn how to read from and write to text files using file streams (ifstream, ofstream).
- To design a menu-driven program that continuously takes choices from the user.
- To implement the five required operations on file-stored records in a clear and structured way.

- To add a simple analysis step that compares average expression between patients and healthy donors, connecting basic programming concepts with a real bioinformatics idea.

System design

The system is designed as a **menu-driven console application**. When the program starts, it repeatedly shows a menu that lets the user choose what operation to perform. The menu includes options for adding, viewing, searching, editing, and deleting records for both diseased and normal samples, as well as an option to exit the program. The loop continues until the user selects Exit, which makes the system easy to use even for beginners.

Two separate text files are used for data storage:

- patient.txt – stores all diseased patient records.
- normal.txt – stores all normal sample records.

Each record is written on one line in the format: ID, Name, Age, GeneName and Expression. This simple format allows reading values using the extraction operator (>>) in the same order they were written. Keeping one record per line also makes it easier for a human to open the file in a text editor and see the data.

The program follows a **modular design.** Instead of writing everything in main(), each operation is implemented as its own function. For example, addPatient() handles adding a diseased patient record, viewNormals() prints all normal records, and similar functions exist for searching, editing, and deleting records. This modular design improves readability and makes it easier to test or change each part of the program independently.

Gene Expression Comparison:

The program reads all gene expression values from patient.txt and normal.txt, computes the average expression for diseased samples and for normal samples, and then compares these averages.

The result is shown in the console as a short message. If the diseased group has a higher average, the program reports that the gene is overly expressed in patients; if the average is lower, it reports under expression; and if both averages are the same, it reports no difference between the two groups.

Implementation details

Data fields

To satisfy the "*at least five variables*" requirement, every record uses:

- **int id** unique ID for the sample.
- **string name** name of the patient or normal donor.
- **int age** age of the person.
- **string geneName** name of the gene being studied.
- **float expression** numerical value of the gene expression level.

These variables are declared locally inside functions and read directly from the keyboard using **cin**. They are then written to the appropriate file using **ofstream** with the insertion operator (<<), separated by spaces.

File handling

The program demonstrates three basic file operations:

- **Appending data:**
  New records are added at the end of the file using ios::app, so old records are not lost.
- **Reading data:**
  To view or search, the program opens the file with ifstream and uses a while loop to read each record in order.
- **Editing/deleting data:**
  The program copies all records into a temporary file, changing or skipping the selected record, then replaces the old file with this new one.

User interface

The user interacts with the system through simple text prompts. The menu shows clear options with numbers,

The user enters an integer choice, and a switch statement in main() calls the corresponding function.

If the user enters an **invalid choice**, a message is displayed and the menu is shown again.

```
===== Gene Expression File Analyzer =====
1. Add Diseased Patient Sample
2. Add Normal Sample
3. View Diseased Patient Samples
4. View Normal Samples
5. Search Diseased Patient Sample
6. Search Normal Sample
7. Edit Diseased Patient Sample
8. Edit Normal Sample
9. Compare Gene Expression (Normal vs Diseased)
10. Delete Diseased Patient Sample
11. Delete Normal Sample
12. Exit
Enter your choice:
```

Code:

1.Functions

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

//  Function prototypes
void addPatient();
void addNormal();

void viewPatients();
void viewNormals();

void searchPatient();
void searchNormal();

void editPatient();
void editNormal();

void deletePatient();
void deleteNormal();

void compareExpression();
```

2. The Menu-driven Code:

```
int main() {
    int choice;
    do {
        cout << "\n===== Gene Expression File Analyzer =====\
        cout << "1. Add Diseased Patient Sample\n";
        cout << "2. Add Normal Sample\n";
        cout << "3. View Diseased Patient Samples\n";
        cout << "4. View Normal Samples\n";
        cout << "5. Search Diseased Patient Sample\n";
        cout << "6. Search Normal Sample\n";
        cout << "7. Edit Diseased Patient Sample\n";
        cout << "8. Edit Normal Sample\n";
        cout << "9. Delete Diseased Patient Sample\n";
        cout << "10. Delete Normal Sample\n";
        cout << "11. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
```

3.Switch Case

```
        cin >> choice;

        switch (choice) {
        case 1:  addPatient();  break;
        case 2:  addNormal();   break;
        case 3:  viewPatients(); break;
        case 4:  viewNormals();  break;
        case 5:  searchPatient(); break;
        case 6:  searchNormal();  break;
        case 7:  editPatient();   break;
        case 8:  editNormal();    break;
        case 9:  deletePatient(); break;
        case 10: deleteNormal();  break;
        case 11: cout << "Program exited successfully.\n"; break;
        default: cout << "Invalid choice!\n";
        }
    } while (choice != 11);
```

4.Do-while loop

```cpp
int choice;
do {
    cout << "\n===== Gene Expression File Analyzer =====\n";
    cout << "1. Add Diseased Patient Sample\n";
    cout << "2. Add Normal Sample\n";
    cout << "3. View Diseased Patient Samples\n";
    cout << "4. View Normal Samples\n";
    cout << "5. Search Diseased Patient Sample\n";
    cout << "6. Search Normal Sample\n";
    cout << "7. Edit Diseased Patient Sample\n";
    cout << "8. Edit Normal Sample\n";
    cout << "9. Delete Diseased Patient Sample\n";
    cout << "10. Delete Normal Sample\n";
    cout << "11. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;
```

```cpp
    switch (choice)
    {
case 1:  addPatient(); break;
case 2:  addNormal(); break;
case 3:  viewPatients(); break;
case 4:  viewNormals(); break;
case 5:  searchPatient(); break;
case 6:  searchNormal(); break;
case 7:  editPatient(); break;
case 8:  editNormal(); break;
case 9:  compareExpression(); break;
case 10: deletePatient(); break;
case 11: deleteNormal(); break;
case 12: cout << "Program exited successfully.\n"; break;
default: cout << "Invalid choice!\n";

    }
} while (choice != 12);

return 0;
}
```

5.Function addPatient

```cpp
// ADD Disease Patient
void addPatient() {
    ofstream file("patient.txt", ios::app);

    int id, age;
    string name, geneName;
    float expression;

    cout << "Enter Patient ID: ";
    cin >> id;
    cout << "Enter Patient Name: ";
    cin >> name;
    cout << "Enter Patient Age: ";
    cin >> age;
    cout << "Enter Gene Name: ";
    cin >> geneName;
    cout << "Enter Gene Expression Level: ";
    cin >> expression;

    file << id << ' ' << name << ' ' << age << ' '
         << geneName << ' ' << expression << '\n';
    file.close();

    cout << "Diseased patient record added.\n";
}
```

6. Function addNormal

```cpp
// ADD Normal
void addNormal() {
    ofstream file("normal.txt", ios::app);

    int id, age;
    string name, geneName;
    float expression;

    cout << "Enter Normal Sample ID: ";
    cin >> id;
    cout << "Enter Donor Name: ";
    cin >> name;
    cout << "Enter Donor Age: ";
    cin >> age;
    cout << "Enter Gene Name: ";
    cin >> geneName;
    cout << "Enter Gene Expression Level: ";
    cin >> expression;

    file << id << ' ' << name << ' ' << age << ' '
         << geneName << ' ' << expression << '\n';
    file.close();

    cout << "Normal sample record added.\n";
}
```

7.Function viewPatients

```
//  VIEW Patient
void viewPatients() {
    ifstream file("patient.txt");
    int id, age;
    string name, geneName;
    float expression;

    cout << "\n--- Diseased Patient Samples ---\n";
    while (file >> id >> name >> age >> geneName >> expression) {
        cout << "ID: " << id
             << " | Name: " << name
             << " | Age: " << age
             << " | Gene: " << geneName
             << " | Expression: " << expression << '\n';
    }
    file.close();
}
```

8. Function viewNormals

```
// View Normal
void viewNormals() {
    ifstream file("normal.txt");
    int id, age;
    string name, geneName;
    float expression;

    cout << "\n--- Normal Samples ---\n";
    while (file >> id >> name >> age >> geneName >> expression) {
        cout << "ID: " << id
             << " | Name: " << name
             << " | Age: " << age
             << " | Gene: " << geneName
             << " | Expression: " << expression << '\n';
    }
    file.close();
}
```

9.Function searchPatient

```
//  SEARCH Pateint
void searchPatient() {
    ifstream file("patient.txt");
    int searchId;
    cout << "Enter Patient ID to search: ";
    cin >> searchId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (file >> id >> name >> age >> geneName >> expression) {
        if (id == searchId) {
            cout << "Record found:\n";
            cout << "ID: " << id
                 << " | Name: " << name
                 << " | Age: " << age
                 << " | Gene: " << geneName
                 << " | Expression: " << expression << '\n';
            found = true;
            break;
        }
    }
    if (!found) {
        cout << "No patient record with this ID.\n";
    }
    file.close();
}
```

10.Function searchNormal

```
// Search Normal
void searchNormal() {
    ifstream file("normal.txt");
    int searchId;
    cout << "Enter Normal Sample ID to search: ";
    cin >> searchId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (file >> id >> name >> age >> geneName >> expression) {
        if (id == searchId) {
            cout << "Record found:\n";
            cout << "ID: " << id
                 << " | Name: " << name
                 << " | Age: " << age
                 << " | Gene: " << geneName
                 << " | Expression: " << expression << '\n';
            found = true;
            break;
        }
    }
    if (!found) {
        cout << "No normal sample record with this ID.\n";
    }
    file.close();
}
```

11.Function editPatient

```
//  EDIT Patient
void editPatient() {
    ifstream in("patient.txt");
    ofstream out("temp.txt");

    int editId;
    cout << "Enter Patient ID to edit: ";
    cin >> editId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (in >> id >> name >> age >> geneName >> expression) {
        if (id == editId) {
            cout << "Enter NEW Name: ";
            cin >> name;
            cout << "Enter NEW Age: ";
            cin >> age;
            cout << "Enter NEW Gene Name: ";
            cin >> geneName;
            cout << "Enter NEW Expression: ";
            cin >> expression;
            found = true;
        }
        out << id << ' ' << name << ' ' << age << ' '
            << geneName << ' ' << expression << '\n';
    }

    in.close();
    out.close();
    remove("patient.txt");
    rename("temp.txt", "patient.txt");

    if (found)
        cout << "Patient record updated.\n";
    else
        cout << "Patient ID not found.\n";
}
```

12.Function editNormal

```cpp
// Edit Normal
void editNormal() {
    ifstream in("normal.txt");
    ofstream out("temp.txt");

    int editId;
    cout << "Enter Normal Sample ID to edit: ";
    cin >> editId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (in >> id >> name >> age >> geneName >> expression) {
        if (id == editId) {
            cout << "Enter NEW Name: ";
            cin >> name;
            cout << "Enter NEW Age: ";
            cin >> age;
            cout << "Enter NEW Gene Name: ";
            cin >> geneName;
            cout << "Enter NEW Expression: ";
            cin >> expression;
            found = true;
        }
        out << id << ' ' << name << ' ' << age << ' '
            << geneName << ' ' << expression << '\n';
    }

    in.close();
    out.close();
    remove("normal.txt");
    rename("temp.txt", "normal.txt");

    if (found)
        cout << "Normal sample record updated.\n";
    else
        cout << "Normal sample ID not found.\n";
}
```

13.Function deletePatient

```cpp
// DELETE Patient
void deletePatient() {
    ifstream in("patient.txt");
    ofstream out("temp.txt");

    int deleteId;
    cout << "Enter Patient ID to delete: ";
    cin >> deleteId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (in >> id >> name >> age >> geneName >> expression) {
        if (id == deleteId) {
            found = true;
            continue;
        }
        out << id << ' ' << name << ' ' << age << ' '
            << geneName << ' ' << expression << '\n';
    }

    in.close();
    out.close();
    remove("patient.txt");
    rename("temp.txt", "patient.txt");

    if (found)
        cout << "Patient record deleted.\n";
    else
        cout << "Patient ID not found.\n";
}
```

14.Function deleteNormal

```
// DELETE Normal
void deleteNormal() {
    ifstream in("normal.txt");
    ofstream out("temp.txt");

    int deleteId;
    cout << "Enter Normal Sample ID to delete: ";
    cin >> deleteId;

    int id, age;
    string name, geneName;
    float expression;
    bool found = false;

    while (in >> id >> name >> age >> geneName >> expression) {
        if (id == deleteId) {
            found = true;
            continue;
        }
        out << id << ' ' << name << ' ' << age << ' '
            << geneName << ' ' << expression << '\n';
    }

    in.close();
    out.close();
    remove("normal.txt");
    rename("temp.txt", "normal.txt");

    if (found)
        cout << "Normal sample record deleted.\n";
    else
        cout << "Normal sample ID not found.\n";
}
```

15. Function Comparison

```
// Comparison of Expression
void compareExpression() {
    string targetGene;
    int id, age;
    string name, gene;
    float expression;

    float pSum = 0, nSum = 0;
    int pCount = 0, nCount = 0;

    ifstream pFile("patient.txt");
    ifstream nFile("normal.txt");

    if (!pFile || !nFile) {
        cout << "Error opening files!" << endl;
        return;
    }

    cout << "Enter Gene Name: ";
    cin >> targetGene;

    while (pFile >> id >> name >> age >> gene >> expression) {
        if (gene == targetGene) {
            pSum = pSum + expression;
            pCount = pCount + 1;
        }
    }
```

```cpp
    while (nFile >> id >> name >> age >> gene >> expression) {
        if (gene == targetGene) {
            nSum = nSum + expression;
            nCount = nCount + 1;
        }
    }

    cout << "--- Results ---" << endl;

    if (pCount > 0 && nCount > 0) {
        float pAvg = pSum / pCount;
        float nAvg = nSum / nCount;

        cout << "Patient Average: " << pAvg << endl;
        cout << "Normal Average: " << nAvg << endl;

        if (pAvg > nAvg) {
            cout << "Result:Expression is Over-expressed in the diseased patient." << endl;
        } else if (pAvg < nAvg) {
            cout << "Result: Expression is Under-expressed in the diseased patient." << endl;
        } else {
            cout << "Result: Same expression." << endl;
        }
    } else {
        cout << "Gene not found in one or both files." << endl;
    }

    pFile.close();
    nFile.close();
}
```

Output:

Menu

```
===== Gene Expression File Analyzer =====
1. Add Diseased Patient Sample
2. Add Normal Sample
3. View Diseased Patient Samples
4. View Normal Samples
5. Search Diseased Patient Sample
6. Search Normal Sample
7. Edit Diseased Patient Sample
8. Edit Normal Sample
9. Compare Gene Expression (Normal vs Diseased)
10. Delete Diseased Patient Sample
11. Delete Normal Sample
12. Exit
Enter your choice: 1
```

Choice 1 : Diseased Patient Sample Entries

```
1. Add Diseased Patient Sample
```

```
Enter your choice: 1
Enter Patient ID: 001
Enter Patient Name: Anaa
Enter Patient Age: 45
Enter Gene Name: BRCA1
Enter Gene Expression Level: 12.5
Diseased patient record added.
```

```
Enter your choice: 1
Enter Patient ID: 002
Enter Patient Name: Sara
Enter Patient Age: 34
Enter Gene Name: BRCA1
Enter Gene Expression Level: 13.5
Diseased patient record added.
```

```
Enter your choice: 1
Enter Patient ID: 003
Enter Patient Name: Einaa
Enter Patient Age: 23
Enter Gene Name: BRCA1
Enter Gene Expression Level: 16.7
Diseased patient record added.
```

Choice 2 : Donor Sample Entries

```
2. Add Normal Sample
```

```
Enter your choice: 2
Enter Normal Sample ID: 001
Enter Donor Name: Akla
Enter Donor Age: 34
Enter Gene Name: BRCA1
Enter Gene Expression Level: 2.38
Normal sample record added.
```

```
Enter your choice: 2
Enter Normal Sample ID: 002
Enter Donor Name: Alikha
Enter Donor Age: 24
Enter Gene Name: BRCA1
Enter Gene Expression Level: 2.00
Normal sample record added.
```

```
Enter your choice: 2
Enter Normal Sample ID: 003
Enter Donor Name: Awisha
Enter Donor Age: 37
Enter Gene Name: BRCA1
Enter Gene Expression Level: 1.89
Normal sample record added.
```

Choice 3:

```
3. View Diseased Patient Samples
```

```
Enter your choice: 3

--- Diseased Patient Samples ---
ID: 234 | Name: Alya | Age: 56 | Gene: BRACA | Expression: 18745
ID: 1 | Name: Anaa | Age: 45 | Gene: BRCA1 | Expression: 12.5
ID: 2 | Name: Sara | Age: 34 | Gene: BRCA1 | Expression: 13.5
ID: 3 | Name: Einaa | Age: 23 | Gene: BRCA1 | Expression: 16.7
```

Choice 4:

```
4. View Normal Samples
```

```
Enter your choice: 4

--- Normal Samples ---
ID: 1 | Name: Akla | Age: 34 | Gene: BRCA1 | Expression: 2.38
ID: 2 | Name: Alikha | Age: 24 | Gene: BRCA1 | Expression: 2
ID: 3 | Name: Awisha | Age: 37 | Gene: BRCA1 | Expression: 1.89
```

Choice 5:

```
5. Search Diseased Patient Sample
```

```
Enter your choice: 5
Enter Patient ID to search: 002
Record found:
ID: 2 | Name: Sara | Age: 34 | Gene: BRCA1 | Expression: 13.5
```

Choice 6:

```
6. Search Normal Sample
```

```
Enter your choice: 6
Enter Normal Sample ID to search: 003
Record found:
ID: 3 | Name: Awisha | Age: 37 | Gene: BRCA1 | Expression: 1.89
```

Choice 7:

```
7. Edit Diseased Patient Sample
```

```
Enter your choice: 7
Enter Patient ID to edit: 003
Enter NEW Name: Aqna
Enter NEW Age: 45
Enter NEW Gene Name: BRCA2
Enter NEW Expression: 23.1
Patient record updated.
```

```
ID: 3 | Name: Einaa | Age: 23 | Gene: BRCA1 | Expression: 16.7
```

Updated:

```
ID: 3 | Name: Aqna | Age: 45 | Gene: BRCA2 | Expression: 23.1
```

Choice 8:

```
8. Edit Normal Sample
```

```
Enter your choice: 8
Enter Normal Sample ID to edit: 003
Enter NEW Name: Anaya
Enter NEW Age: 23
Enter NEW Gene Name: BRCA2
Enter NEW Expression: 2.5
Normal sample record updated.
```

```
ID: 3 | Name: Awisha | Age: 37 | Gene: BRCA1 | Expression: 1.89
```

Updated:

```
ID: 3 | Name: Anaya | Age: 23 | Gene: BRCA2 | Expression: 2.5
```

Choice 9:

```
Comparison of the Gene Expression:
Enter Gene Name: BRCA1
```

```
Enter your choice: 9
Comparison of the Gene Expression:
Enter Gene Name: BRCA1
--- Results ---
Patient Average: 12.025
Normal Average: 2.38
Result:Expression is Over-expressed in the diseased patient.
```

Choice 10: `10. Delete Diseased Patient Sample`

Updated:

```
Enter your choice: 3

--- Diseased Patient Samples ---
ID: 2 | Name: Sara | Age: 34 | Gene: BRCA1 | Expression: 13.5
ID: 3 | Name: Aqna | Age: 45 | Gene: BRCA2 | Expression: 23.1
```

Choice 11: `11. Delete Normal Sample`

```
Enter your choice: 4

--- Normal Samples ---
ID: 1 | Name: Akla | Age: 34 | Gene: BRCA1 | Expression: 2.38
ID: 2 | Name: Alikha | Age: 24 | Gene: BRCA1 | Expression: 2
ID: 3 | Name: Awisha | Age: 37 | Gene: BRCA1 | Expression: 1.89
```

Updated:

```
Enter your choice: 4

--- Normal Samples ---
ID: 1 | Name: Akla | Age: 34 | Gene: BRCA1 | Expression: 2.38
ID: 3 | Name: Anaya | Age: 23 | Gene: BRCA2 | Expression: 2.5
```

Choice 12: `12. Exit`

```
Enter your choice: 12
Program exited successfully.
```

Conclusion

The **Gene Expression File Analyzer** successfully implements all requirements of the project. Each record uses at least five variables, records are permanently stored in text files, and the system supports adding, viewing, searching, editing, and deleting data for diseased and normal samples through a friendly menu-based interface.

In addition to basic file handling practice, the project demonstrates how programming can be used to explore a real biological question by comparing average gene expression between patients and healthy donors.

Files:

patient - Notepad

File Edit Format View Help

```
2 Sara 34 BRCA1 13.5
3 Aqna 45 BRCA2 23.1
1 Ashna 23 BRCA1 9
```

normal - Notepad

File Edit Format View Help

```
1 Aqla 34 BRCA1 2.38
3 Anaya 23 BRCA2 2.5
```

Data directly added in the files.